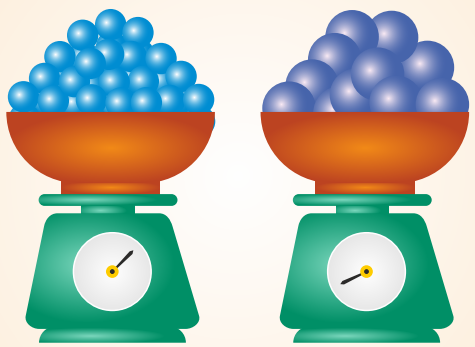
ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

path to success

PHYSICAL CHEMISTRY

JEE (MAIN+ADVANCED)

NURTURE COURSE



Exercise

Mole Concept & Eudiometry

(English Medium)

EXERCISE # S- I

ATOMIC MASS , MOLECULAR MASS, MOLE

1. Find :

(i) No. of moles of Cu atom in 10^{23} atoms of Cu. **MC0001**

(ii) Mass of 200 $^{16}_{8}O$ atoms in amu **MC0001**

(iii) Mass of 100 atoms of $^{14}_{7}N$ is $y \times 10^{-22}$ in gm ,then value of y is (1 amu = 1.67×10^{-24} gm) **MC0001**

(iv) No. of molecules & atoms in 54 gm H_2O is $y \times 10^{23}$ and $z \times 10^{23}$ respectively then value of y & z is ($N_A = 6.022 \times 10^{23}$) **MC0002**

(v) No. of molecules in 88 gm CO_2 is $y \times 10^{23}$, then value of y is ($N_A = 6.022 \times 10^{23}$) **MC0002**

2. If mass of one ^{12}C atom is $y \times 10^{-23}$ gm, then value of y is ? **MC0003**

3. Calculate mass (in gm) of O atoms in 6 gm CH_3COOH ? **MC0004**

4. Calculate mass of water (in gm) present in 499 gm $CuSO_4.5H_2O$?
(Atomic mass : Cu = 63.5, S = 32, O = 16, H = 1) **MC0005**

5. What mass (in gm) of $Na_2SO_4.7H_2O$ contains exactly 6.022×10^{22} atoms of oxygen ?
($N_A = 6.022 \times 10^{23}$) **MC0006**

6. The weight (in gram) of pure potash Alum ($K_2SO_4.Al_2(SO_4)_3.24H_2O$) which contains 6.4 gm oxygen is. (Atomic weight of K = 39, S = 32, Al = 27) **MC0007**

7. The Kohinoor diamond was the largest diamond ever found. How many moles of carbon atom were peresent in it, if it is weigh 3300 carat. [Given: 1 carat = 200 mg] **MC0008**

8. Calculate volume (in litre) of H_2 gas kept at STP if it contains as many H atoms as in 98 gm H_3PO_4.
[Atomic mass of P = 31] **MC0009**

9. 80gm of SO_x gas occupies 14 litre at 2atm & 273K. The value of x is : **MC0010**

10. 40 mg of gaseous substance (X_2) occupies 4.8 mL of volume at 1 atm and 27°C. Atomic mass of element X is : (R : 0.082 atm L/mole-K) **MC0011**

STOICHIOMETRY

11. How many gm of HCl is needed for complete reaction with 21.75 gm MnO_2 ?
(Mn = 55, Cl = 35.5)

$HCl + MnO_2 \longrightarrow MnCl_2 + H_2O + Cl_2$ **MC0012**

12. Nitric acid is manufactured by the Ostwald process, in which nitrogen dioxide reacts with water.

$$3\ NO_2\ (g) + H_2O\ (l) \rightarrow 2\ HNO_3\ (aq) + NO\ (g)$$

How many grams of nitrogen dioxide are required in this reaction to produce 25.2 gm HNO_3 ? **MC0013**

13. Flourine reacts with uranium to produce uranium hexafluoride, UF_6, as represented by this equation

$$U(s) + 3F_2(g) \rightarrow UF_6(g)$$

If no. of fluorine molecules are required to produce 7.04 mg of uranium hexafluoride, UF_6, from an excess of uranium is $y \times 10^{19}$ then value of y is ? The molar mass of UF_6 is 352 gm/mol.
($N_A = 6.022 \times 10^{23}$) **MC0014**

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14. What total volume, in litre at 627°C and 82.1 atm, could be formed by the decomposition of 16 gm of NH_4NO_3 ? Reaction : $2\ NH_4NO_3 \rightarrow 2N_2 + O_2 + 4H_2O_{(g)}$. (R : 0.0821 atm L/mole-K) **MC0015**

15. Calculate mass of phosphoric acid (in gm) required to obtain 53.4g pyrophosphoric acid.

$2H_3PO_4 \rightarrow H_4P_2O_7 + H_2O$ **MC0016**

LIMITING REACTANT

16. Carbon reacts with chlorine to form CCl_4. 36 gm of carbon was mixed with 142 g of Cl_2. If ratio of mass of CCl_4 produced and the remaining mass of excess reactant is y : 1, then value of y is **MC0017**

17. Potassium superoxide, KO_2, is used in rebreathing gas masks to generate oxygen :

$$KO_2(s) + H_2O(l) \rightarrow KOH\ (s) + O_2\ (g)$$

If a reaction vessel contains 0.158 mol KO_2 and 0.10 mol H_2O, how many moles of O_2 can be produced ? **MC0018**

18. A chemist wants to prepare diborane (B_2H_6) by the reaction

$6\ LiH + 8BF_3 \longrightarrow 6Li\ BF_4 + B_2H_6$

If he starts with 2.0 moles each of LiH & BF_3. How many moles of B_2H_6 can be prepared. **MC0019**

19. Sulphuric acid is produced when sulphur dioxide reacts with oxygen and water in the presence of a catalyst : $2SO_2(g) + O_2\ (g) + 2\ H_2O(l) \rightarrow 2\ H_2SO_4$. If 5.6 mol of SO_2 reacts with 4.8 mol of O_2 and a large excess of water, what is the maximum number of moles of H_2SO_4 that can be obtained ? **MC0020**

PROBLEMS RELATED WITH MIXTURE

20. One gram of an alloy of aluminium and magnesium when heated with excess of dil. HCl forms magnesium chloride, aluminium chloride and hydrogen. The evolved hydrogen collected at 0°C has a volume of 1.12 litres at 1 atm pressure. Calculate the composition of (% by mass) of the alloy.

(Al = 27, Mg = 24) **MC0021**

21. A sample containing only $CaCO_3$ and $MgCO_3$ is ignited to CaO and MgO. The mixture of oxides produced weight exactly half as much as the original sample. Calculate the percentages of $CaCO_3$ and $MgCO_3$ (by mass) in the sample. **MC0022**

22. Determine the percentage composition (by mass) of a mixture of anhydrous sodium carbonate and sodium bicarbonate from the following data:

wt. of the mixture taken = 2g

Loss in weight on heating = 0.124 gm. **MC0023**

23. A sample of mixture of $CaCl_2$ and NaCl weighing 2.22 gm was treated to precipitate all the Ca as $CaCO_3$ which was then heated and quantitatively converted to 0.84 gm of CaO. Calculate mass fraction of $CaCl_2$ in the mixture. **MC0024**

24. When 4 gm of a mixture of $NaHCO_3$ and NaCl is heated, 0.66 gm CO_2 gas is evolved. If ratio of the percentage composition (by mass) of the $NaHCO_3$ and NaCl is y : 1 then value of y is. **MC0025**

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PERCENTAGE YIELD, DEGREE OF DISSOCIATION , PERCENTAGE

25. A power company burns approximately 500 tons of coal per day to produce electricity. If the sulphur content of the coal is 1.20 % by weight, how many tons SO_2 are dumped into the atmosphere per hour ? **MC0026**

26. Calculate the percent loss in weight after complete decomposition of a pure sample of potassium chlorate. (K = 39, Cl = 35.5)

$$KClO_3(s) \longrightarrow KCl(s) + O_2(g)$$ **MC0027**

27. A sample of calcium carbonate is 80% pure, 25 gm of this sample is treated with excess of HCl.How much volume (in litre) of CO_2 will be obtained at 1 atm & 273 K? **MC0028**

28. Cyclohexanol is dehydrated to cyclohexene on heating with conc. H_2SO_4. If the yield of this reaction is 75%, how much cyclohexene (in gm) will be obtained from 100 g of cyclohexanol ?

$$C_6H_{12}O \xrightarrow{\text{con. } H_2SO_4} C_6H_{10}$$ **MC0029**

29. If the yield of chloroform obtainable from acetone and bleaching powder is 58%. What is the weight of acetone (in gm) required for producing 239 mg of chloroform ?

$2CH_3COCH_3 + 6CaOCl_2 \rightarrow Ca(CH_3COO)_2 + 2CHCl_3 + 3CaCl_2 + 2Ca(OH)_2$ **MC0030**

30. Calculate % yield of the reaction if 200g $KHCO_3$ produces 22g of CO_2 upon strong heating. (K = 39) **MC0031**

31. The vapour density of a sample of N_2O_4 gas is 35. What percent of N_2O_4 molecules are dissociated into NO_2 ? . **MC0032**

32. If a sample of pure SO_3 gas is heated to 600ºC, it dissociates into SO_2 and O_2 gases upto 50%. If the average molar mass of the final sample is M_{avg} find value of $\left(\frac{M_{av}}{100}\right)$. **MC0033**

33. When silent electric discharge is passed through O_2 gas, it converts into O_3. If the density of final sample is 20 times the density of hydrogen gas under similar conditions, calculate the mass percent of O_2 in the final sample. **MC0034**

34. When acetylene (C_2H_2) gas is passed through red hot iron tube, it trimerises into benzene (C_6H_6) vapours. If the average molar mass of vapours coming out through the tube is 50, calculate the degree of trimerisation of acetylene. **MC0035**

SEQUENTIAL & PARALLEL REACTIONS

35. $Br_2(l)$ reacts with $Cl_2(g)$ to form BrCl and $BrCl_3$, simultaneously. How many moles of $Cl_2(g)$ reacts completely with 0.03 moles of $Br_2(l)$ to form BrCl and $BrCl_3$ in 5 : 1 molar ratio **MC0036**

36. When 80 gm CH_4 is burnt , CO and CO_2 gases are formed in 1 : 4 mole ratio. If the mass of O_2 gas used in combustion is w gm then find value of (w/100). **MC0037**

37. Sulphur trioxide may be prepared by the following two reactions :

$$S_8 + 8O_2(g) \rightarrow 8SO_2(g)$$
$$2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$$

If amount of SO_3 will be produced from 1 mol of S_8 is w gm, then value of w/100 is. **MC0038**

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38. Potassium superoxide, KO_2, is utilised in closed system breathing apparatus. Exhaled air contains CO_2 and H_2O, both of which are removed and the removal of water generates oxygen for breathing by the reaction

$4KO_2(s) + 2H_2O(l) \rightarrow 3O_2(g) + 4KOH(s)$

The potassium hydroxide removes CO_2 from the apparatus by the reaction :

$KOH(s) + CO_2(g) \rightarrow KHCO_3(s)$

(a) What mass of KO_2 (in gm) generates 0.24 gm of oxygen ?

(b) What mass of CO_2 (in gm) can be removed from the apparatus by 100 gm of KO_2 ? (K = 39)

MC0039

PRINCIPLE OF ATOM CONSERVATION

39. In a determination of P, an aqueous solution of NaH_2PO_4 is treated with a mixture of ammonium and magnesium ions to precipitate magnesium ammonium phosphate $Mg(NH_4)PO_4.\ 6H_2O$. This is heated and decomposed to magnesium pyrophosphate, $Mg_2P_2O_7$ which is weighed. A solution of NaH_2PO_4 yielded 1.11 g of $Mg_2P_2O_7$. What weight of NaH_2PO_4 (in gm) was present originally ? (P = 31)

MC0040

40. 6 gm nitrogen on successive reaction with different compounds gets finally converted into 30 gm $[Cr(NH_3)_xBr_2]$ Value of x is [Atomic mass : Cr = 52, Br = 80] **MC0041**

41. A 5.00 gm sample of a natural gas, consisting of methane (CH_4), and ethylene, (C_2H_4) was burned in excess oxygen, yielding $\frac{44}{3}$ gm of CO_2 and some H_2O as products. What mole percent of the sample was ethylene? **MC0042**

42. All carbon atom present in $KH_3(C_2O_4)_2.2H_2O$ weighing 7.62 gm is converted to CO_2. How many gm of CO_2 were obtained. (K = 39) **MC0043**

PERCENTAGE COMPOSITION, EMPERICAL AND MOLECULAR FORMULA

43. Haemoglobin contains 0.25% iron by mass. The molecular mass of of Haemoglobin is 89600. The number of iron atoms per molecule of Haemoglobin (Atomic mass of Fe = 56) - **MC0044**

44. 1.6 g of an organic compound containing sulphur, when treated with series of reagents, produces H_2SO_4 which on reaction with $BaCl_2$ produces 0.233 g of $BaSO_4$. Calculate % by mass of S in the organic compound. **(Given : Atomic weight of Ba = 137)** **MC0045**

45. When 2.0 gm of an organic compound is burnt completely, 150 ml N_2 gas at 27°C and 0.821 atm is obtained. The mass fraction of nitrogen in the compound is.

(R = 0.0821 atm L/mole - K) **MC0046**

46. A polystyrene of formula $Br_3C_6H_2(C_8H_8)_n$ was prepared by heating styrene with tribromobenzyl peroxide in the absence of air. It was found to contain 10.48% bromine by weight. Find the value of n. (Br = 80). **MC0047**

47. A moth repellent has the composition 49% C, 2.7% H and 48.3% Cl. Its molecular weight is 147 gm. Determine its molecular formula **MC0048**

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48. 0.5 g of NaOH is required by 0.4 gm of a polybasic acid H_nA (Molecular weight = 96gm) for complete neutralization. Value of 'n' would be : (Assume all H atom are replaced)– **MC0049**

49. The empirical formula of a compounds is CH_2O. 0.25 mole of this compound contains 1 gm hydrogen. The molecular formula of compound is - **MC0050**

50. A compound has 62 % carbon, 10.4 % hydrogen and 27.6 % oxygen. If molar mass of compound is 58, if ratio of no. of C atom to H-atom is y : 1 then value of y is. **MC0051**

EUDIOMETRY

51. What volume (in ml) of $O_2(g)$ is needed for complete combustion of 40 ml ethane gas (C_2H_6) ? **MC0052**

52. 10 ml of CO is mixed with 25 ml air having 20% O_2 by volume. What would be the final volume (in ml) if none of CO and O_2 is left after the reaction? **MC0053**

53. Calculate the volume (in ml) of CO_2 evolved by the combustion of 50 ml of a mixture containing 40% C_2H_4 and 60% CH_4 (by volume) **MC0054**

54. 10 moles of a mixture of CO (g) and $CH_4(g)$ was mixed with 22 moles of O_2 gas and subjected to sparking. Find the moles of gas absorbed when the residual gases are passed through alc. KOH. **MC0055**

55. 60 ml of a mixture of nitrous oxide and nitric oxide was exploded with excess of hydrogen. If 38 ml of N_2 was formed, calculate % volume of NO gas in the mixture. **MC0056**

56. When 100 ml of a $O_2 - O_3$ mixture was passed through turpentine, there was reduction of volume by 20 ml. If 100 ml of such a mixture is heated, what will be the increase in volume (in ml)? **MC0057**

57. 10 ml of an oxide of nitrogen produce 20 ml NO_2 and 5 ml O_2 on complete decomposition. The oxide of nitrogen is- **MC0058**

58. A gaseous alkane is exploded with O_2. The volume of O_2 required for complete combustion and the volume of CO_2 formed after combustion are in 7 : 4 ratio. What is the molecular formula of alkane ? **MC0059**

59. When a certain quantity of oxygen was ozonised in a suitable apparatus, the volume decreased by 4 ml. On addition of turpentine the volume further decreased by 8 ml. All volumes were measured at the same temperature and pressure. From these data, if formula of ozone is O_x then find x. **MC0060**

60. A 20 ml mixture of C_2H_4 and C_2H_2 undergoes sparking in gas eudiometer with just sufficient amount of O_2 and shows contraction of 37.5 ml. The volume (in ml) of C_2H_2 in the mixture is. **MC0061**

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EXERCISE # S-II

MOLE

1. Two substance P_4 & O_2 are allowed to react completely to form mixture of P_4O_6 & P_4O_{10} leaving none of the reactants. Using this information calculate the moles of P_4O_6 and P_4O_{10} in the final mixture when the following amounts of P_4 & O_2 are taken.

$$P_4 + 3O_2 \longrightarrow P_4O_6$$
$$P_4 + 5O_2 \longrightarrow P_4O_{10}$$

(i) If 1 mole P_4 & 4 mole of O_2

(ii) If 3 mole P_4 & 11 mole of O_2

(iii) If 3 mole P_4 & 13 mole of O_2 **MC0062**

2. By the reaction of carbon and oxygen, a mixture of CO and CO_2 is obtained. What is the mass percent of CO in of the mixture obtained when 20 grams of O_2 reacts with 12 grams of carbon ?

MC0063

3. Nitrogen (N), phosporus (P), and potassium (K) are the main nutrients in plant fertilizers. According to an industry convention, the numbers on the label refer to the mass % of N, P_2O_5, and K_2O, in that order. If the N : P : K ratio of a 28: 11.75 : 11.75 fertilizer in terms of moles of each elements, and express it as x : y : 1. Find value of y (P = 31, K = 39) **MC0064**

4 A 10 g sample of a mixture of calcium chloride and sodium chloride is treated with Na_2CO_3 to precipitate calcium as calcium carbonate. This $CaCO_3$ is heated to convert all the calcium to CaO and the final mass of CaO is 1.12gm. Calculate % by mass of NaCl in the original mixture. **MC0065**

5. A mixture of Ferric oxide (**Fe_2O_3**) and **Al** is used as a solid rocket fuel which reacts to give **Al_2O_3** and **Fe**. No other reactants and products are involved. On complete reaction of 1 mole of **Fe_2O_3**, 200 units of energy is released.

(a) Write a balance reaction representing the above change.

(b) If the ratio of masses of Fe_2O_3 and Al taken so that maximum energy per unit mass of fuel is released is y : 1 then value of y is

(c) What would be energy released if 16 kg of Fe_2O_3 reacts with 2.7 kg of Al. **MC0066**

6. 5.33 mg of salt $[Cr(H_2O)_5Cl].Cl_2.\ H_2O$ is treated with excess of $AgNO_3$(aq.), then mass (in mg) of AgCl precipitate obtained will be :

Given : [Cr = 52, Cl = 35.5, Ag = 108] **MC0067**

7. Mass % of oxygen in monovalent metal carbonate is 48%. If the number of atoms of metal present in 5mg of this metal carbonate sample is $y \times 10^{19}$ then value of y is ($N_A = 6.022 \times 10^{23}$). **MC0068**

8. The formula of compound composed of A & B which is given by A_xB_y, it is strongly heated in oxygen as per reaction-

$A_xB_y + O_2 \rightarrow AO$ + Oxide of B

If 2.5gm of A_xB_y on oxidation gives 3gm oxide of A, If ratio of x : y is z : 1 then value of z is.

[Atomic mass of A = 24 & B = 14] **MC0069**

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9. Calculate the maximum mass of $CaCl_2$ produced when 2.4×10^{24} atoms of calcium is taken with 96 litre of Cl_2 gas at 380 mm pressure and at 27ºC.
[R = 0.08 atm L/mole-K & $N_A = 6 \times 10^{23}$] **MC0070**

10. $P_4S_3 + 8O_2 \longrightarrow P_4O_{10} + 3SO_2$
Calculate mass in gm of P_4S_3 is required to produce at least 9.6 gm of each product. (P = 31, S =32)
MC0071

11. Consider the given reaction
$H_4P_2O_7 + 2NaOH \rightarrow Na_2H_2P_2O_7 + 2H_2O$
If 534 gm of $H_4P_2O_7$ is reacted with 3.0×10^{24} formula units of NaOH, then the total number of moles of H_2O produced is ($N_A = 6 \times 10^{23}$) (P = 31) **MC0072**

Comprehension based on "Law of Conservation of Mass" (12 & 13)

It states that matter can neither be created nor destroyed.

This law was put forth by Antoine Lavoisier in 1789. He performed careful experimental studies for combustion reactions for reaching to the above conclusion. This law formed the basis for several later developments in chemistry. Infact, this was the result of exact measurement of masses of reactants and products, and carefully planned experiments performed by Lavoisier.

12 What weight (in gm) of silver nitrate will react with 0.585 gm NaCl to produce 1.435 gm AgCl and 0.85 gm $NaNO_3$ **MC0073**

13 6.3 gm sodium bicarbonate is added to 15 gm acetic acid solution. CO_2 formed is allowed to escape. The weight of the solution left is 18 gm. What is the mass (in gm) of CO_2 formed. **MC0073**

Comprehension based on "Law of Definite Proportions" (14 & 15)

This law was given by, a French chemist, Joseph Proust. He stated that a given compound always contains exactly the same proportion of elements by weight.
Proust worked with two samples of cupric carbonate —one of which was of natural origin and the other was synthetic one. He found that the composition of elements present in it was same for both the samples as shown below :

	% of copper	% of oxygen	% of carbon
Natural Sample	51.35	9.74	38.91
Synthetic Sample	51.35	9.74	38.91

Thus, irrespective of the source, a given compound always contains same elements in the same proportion. The validity of this law has been confirmed by various experiments. It is sometimes also referred to as **Law of constant composition**.

Limitation :

The law is not applicable if the compound is formed from different isotopes of an element. The two isotopes of carbon C-12 and C-14 form carbondioxide $C^{12}O_2$ and $C^{14}O_2$. The ratio of C : O is 12 : 32 and 14 : 32 respectively. It is not a constant ratio.

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14 0.5 gm silver is dissolved in excess of nitric acid. This solution is treated with excess of NaCl solution when 0.66 gm AgCl is formed. One gram metallic silver wire is heated in dry Cl_2, 1.32 gm AgCl is formed. Show that these data confirm the law of constant proportion. **MC0074**

15. 6.488 gm lead reacts with 1.002 gm oxygen to form an oxide. This oxide is also obtained by heating $Pb(NO_3)_2$, It is found that % of lead in this oxide is 86.62. Show that these date illustrate the law of definite proportions. **MC0075**

Comprehension based on "Law of Multiple Proportions" (16 to 17)

This law was proposed by Dalton in 1803. According to this law, if two elements combine to form more than one compound, the masses of one element that combine with a fixed mass of the other element, are in the ratio of small whole numbers.

For example, hydrogen combines with oxygen to form two compounds, namely, water and hydrogen peroxide.

Hydrogen + Oxygen → Water
2g 16g 18g
Hydrogen + Oxygen → Hydrogen Peroxide
2g 32g 34g

Here, the masses of oxygen (i.e. 16 gm and 32 gm) which combine with a fixed mass of hydrogen (2 gm) bear a simple ratio, i.e. 16:32 or 1: 2.

16. An element forms two oxides. In one oxide, one gram of the oxide contains 0.5 gm of the element. In another oxide, 4 gm of the oxide contains 0.8 gm of the element. Show that these data confirm the law of multiple proportion. **MC0076**

17. 0.11gm of an oxide of nitrogen gives 56 mL N_2 at 273 K and 1atm. 0.15 gm of another oxide of nitrogen gives 56 mL N_2 at 1atm, 273K. Show that these data confirm the law of multiple proportion. **MC0077**

18. 10 ml of a mixture of CH_4, C_2H_4 and CO_2 were exploded with excess of air. After explosion and further cooling, there was contraction of 17 ml and after treatment with KOH, there was further reduction of 14 ml. What is the composition of the mixture (in ml)? **MC0078**

19. 40 ml of a mixture of C_2H_2 and CO is mixed with 100 ml of O_2 gas and the mixture is exploded. The residual gases occupied 104 ml and when these are passed through KOH solution, the volume becomes 48 ml. All the volume are at same temperature and pressure. If ratio of volume of C_2H_2 & CO is y : 1, then value of y is **MC0079**

20. 10 mL of gaseous organic compound containing C, H and O only, was mixed with 100 mL of O_2 and exploded under identical conditions and then cooled. The volume left after cooling was 90 mL. On treatment with KOH a contraction of 20 mL was observed. If vapour density of compound is 23, if molecular formula of the compound is $C_xH_yO_z$, then find (x + y + z). **MC0080**

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EXERCISE # O-I

SINGLE CORRECT :

ATOMIC MASS, MOLECULAR MASS MOLE

1. One atomic mass unit in kilogram is

(A) $1/N_A$ (B) $12 / N_A$ (C) $1/1000\ N_A$ (D) $1000 / N_A$ **MC0081**

2. A gaseous mixture contains CO_2 (g) and N_2O (g) in a 2 : 5 ratio by mass. The ratio of the number of molecules of CO_2 (g) and N_2O (g) is

(A) 5 : 2 (B) 2 : 5 (C) 1 : 2 (D) 5 : 4 **MC0082**

3. Which of the following contain largest number of carbon atoms?

(A) 15 gm ethane, C_2H_6 (B) 40.2 gm sodium oxalate, $Na_2C_2O_4$

(C) 72 gm glucose, $C_6H_{12}O_6$ (D) 35 gm pentene, C_5H_{10} **MC0083**

4. The number of hydrogen atoms in 0.9 gm glucose, $C_6H_{12}O_6$, is same as

(A) 0.048 gm hydrazine, N_2H_4 (B) 0.17 gm ammonia, NH_3

(C) 0.30 gm ethane, C_2H_6 (D) 0.03 gm hydrogen, H_2 **MC0084**

5. Ethanol, C_2H_5OH, is the substance commonly called alcohol. The density of liquid alcohol is 0.7893 g/ml at 293 K. If 1.2 mole of ethanol are needed for a particular experiment, what volume of ethanol should be measured out?

(A) 55 ml (B) 58 ml (C) 70 ml (D) 79 ml **MC0085**

6. An iodized salt contains 0.5 % of NaI. A person consumes 3 gm of salt everyday. The number of iodide ions going into his body everyday is (I = 127)

(A) 10^{-4} (B) 6.02×10^{-4} (C) 6.02×10^{19} (D) 6.02×10^{23} **MC0086**

7. The percentage by mole of NO_2 in a mixture of NO_2(g) and NO(g) having average molecular mass 34 is :

(A) 25% (B) 20% (C) 40% (D) 75% **MC0087**

8. The number of carbon atoms present in a signature, if a signature written by carbon pencil, weighing 1.2×10^{-3} g is

(A) 12.04×10^{20} (B) 6.02×10^{19} (C) 3.01×10^{19} (D) 6.02×10^{20} **MC0088**

9. The average atomic mass of a mixture containing 79 mole % of ^{24}Mg and remaining 21 mole % of ^{25}Mg and ^{26}Mg , is 24.31. **% mole** of ^{26}Mg is

(A) 5 (B) 20 (C) 10 (D) 15 **MC0089**

10. How many litres of oxygen at 1atm & 273K will be required to burn completely 2.2 g of propane (C_3H_8)

(A) 11.2 L (B) 22.4 L (C) 5.6 L (D) 44.8 L **MC0090**

11. If 1/2 moles of oxygen combine with aluminium to form Al_2O_3 then weight of Aluminium metal used in the reaction is (Al= 27)

(A) 27 g (B) 18 g (C) 54 g (D) 40.5 g **MC0091**

12. Volume of CO_2 obtained at STP by the complete decomposition of 9.85 g $BaCO_3$ is (Ba = 137)

(A) 2.24 lit (B) 1.12 lit (C) 1.135 lit (D) 2.27 lit **MC0092**

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13. The drain cleaner, **Drainex** contains small bits of aluminium (Al = 27) which reacts with caustic soda to produce dihydrogen. What is the volume (in ml) of dihydrogen at 27ºC and 1.013 bar that is produced when 0.27 gm of aluminium reacts :

$2Al + 2NaOH + 2H_2O \rightarrow 2NaAlO_2 + 3H_2$

(A) 0.369 (B) 369.0 (C) 246.0 (D) 540.0 **MC0093**

14. Volume of O_2 obtained at 2 atm & 546K, by the complete decomposition of 8.5 g $NaNO_3$ is

$2NaNO_3 \rightarrow 2NaNO_2 + O_2$

(A) 2.24 lit (B) 1.12 lit (C) 0.84 lit (D) 0.56 lit **MC0094**

LIMITING REAGENT

15. The minimum mass of mixture of A_2 and B_4 required to produce at least 1 kg of each product is :
(Given At. mass of 'A' = 10 ; At. mass of 'B' = 120)

$$5A_2 + 2B_4 \longrightarrow 2AB_2 + 4A_2B$$

(A) 2120 gm (B) 1060 gm (C) 560 gm (D) 1660 gm **MC0095**

16. The mass of CO_2 produced from 620 gm mixture of $C_2H_4O_2$ & O_2, prepared to produce maximum energy is (Combustion reaction is exothermic)

(A) 413.33 gm (B) 593.04 gm (C) 440 gm (D) 320 gm **MC0096**

17. The mass of P_4O_{10} produced if 440 gm of P_4S_3 is mixed with 384 gm of O_2 is

$P_4S_3 + O_2 \longrightarrow P_4O_{10} + SO_2$

(A) 568 gm (B) 426 gm (C) 284 gm (D) 396 gm **MC0097**

18. The mass of Mg_3N_2 produced if 48 gm of Mg metal is reacted with 34 gm NH_3 gas is

$$Mg + NH_3 \longrightarrow Mg_3N_2 + H_2$$

(A) $\frac{200}{3}$ gm (B) $\frac{100}{3}$ gm (C) $\frac{400}{3}$ gm (D) $\frac{150}{3}$ gm **MC0098**

PROBLEM RELATED WITH MIXTURE

19. An ideal gaseous mixture of ethane (C_2H_6) and ethene (C_2H_4) occupies 28 litre at 1atm, 0°C. The mixture reacts completely with 128 gm O_2 to produce CO_2 and H_2O. Mole fraction of C_2H_6 in the mixture is–

(A) 0.6 (B) 0.4 (C) 0.5 (D) 0.8 **MC0099**

20. 280 g of a mixture containing CH_4 and C_2H_6 in 5 : 2 molar ratio is burnt in presence of excess of oxygen. Calculate total moles of CO_2 produced.

(A) 9 (B) 18 (C) 7 (D) 12 **MC0100**

21. Mixture of $MgCO_3$ & $NaHCO_3$ on strong heating gives CO_2 & H_2O in 3 : 1 mole ratio. The weight % of $NaHCO_3$ present in the mixture is:

(A) 30% (B) 80% (C) 40% (D) 50% **MC0101**

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PERCENTAGE YIELD, DEGREE OF DISSOCIATION, PERCENTAGE PURITY

22. A metal carbonate decomposes according to following reaction

$$M_2CO_3(s) \longrightarrow M_2O(s) + CO_2(g)$$

Percentage loss in mass on complete decomposition of $M_2CO_3(s)$
(Atomic mass of M = 102)

(A) $\frac{100}{3}\%$ (B) $\frac{50}{3}\%$ (C) $\frac{25}{3}\%$ (D) 15% **MC0102**

23. 90 gm mixture of H_2 and O_2 is taken in stoichiometric ratio and gives H_2O with 50% yield. The produced mass of H_2O (in gm) is :

(A) 45 gm (B) 36 gm (C) 20 gm (D) 90 gm **MC0103**

24. An impure sample of $CaCO_3$ contains 38% of Ca. The percentage of impurity present in the sample is :

(A) 5% (B) 95% (C) 10% (D) 2.5% **MC0104**

25. The vapour density of sample of partially decomposed cyclobutane (C_4H_8) gas is 20. The degree of dissociation of C_4H_8 into C_2H_4 gas is -

(A) 0.25 (B) 0.50 (C) 0.60 (D) 0.40 **MC0105**

26. A sample of NH_3 gas is 20% dissociated into N_2 and H_2 gases. The mass ratio of N_2 and NH_3 gases in the final sample is -

(A) $\frac{7}{34}$ (B) $\frac{7}{17}$ (C) $\frac{14}{17}$ (D) $\frac{21}{17}$ **MC0106**

27. The density of a sample of SO_3 gas is 2.5 g/L at 0ºC and 1 atm. It's degree of dissociation into SO_2 and O_2 gases is -

(A) $\frac{6}{7}$ (B) $\frac{1}{7}$ (C) $\frac{3}{7}$ (D) $\frac{5}{7}$ **MC0107**

SEQUENTIAL AND PARALLEL REACTIONS

28. Polyethene can be prepared by CaC_2 by the following sequence of reactions.

$CaC_2 + H_2O \rightarrow CaO + C_2H_2$

$C_2H_2 + H_2 \rightarrow C_2H_4$

$nC_2H_4 \rightarrow (C_2H_4)_n$

(Polyethene)

The mass in kg of polyethene that can be prepared by 20 kg CaC_2.

(A) 4.1 kg (B) 8.75 kg (C) 3.78 kg (D) 10 kg **MC0108**

29. 25.4 gm of iodine and 14.2 gm of chlorine are made to react completely to yield a mixture of ICl and ICl_3. Ratio of moles of ICl & ICl_3 formed is (Atomic mass : I = 127, Cl = 35.5)

(A) 1 : 1 (B) 1 : 2 (C) 1 : 3 (D) 2 : 3 **MC0109**

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30. One commercial system removes SO_2 emission from smoke at 95ºC by the following set of reactions-

$SO_2(g) + Cl_2(g) \rightarrow SO_2Cl_2(g)$

$SO_2Cl_2 + 2H_2O \rightarrow H_2SO_4 + 2HCl$

$H_2SO_4 + Ca(OH)_2 \rightarrow CaSO_4 + 2H_2O$

Assuming the process to be overall 95% efficient. How many moles of $CaSO_4$ may be produced from 128 gm SO_2. [Ca = 40, S- 32, O-16]

(A) 1.9 moles (B) 2 mol (C) 3.8 mol (D) 0.95 mol **MC0110**

31. Equal masses of $KClO_3$ undergoes different reaction in two different container :

(i) $2KClO_3 \longrightarrow 2KCl + 3O_2$

(ii) $4KClO_3 \longrightarrow KCl + 3KClO_4$

Mass ratio of the KCl produced in respective reaction is x : 1. Value of '**x**' will be.

(A) 4 (B) 2 (C) 0.25 (D) 3 **MC0111**

PERCENTAGE COMPOSITION, EMPIRICAL AND MOLECULAR FORMULA

32. A compound contains 10^{-2} % of phosphorous. If atomic mass of phosphorus is 31, the molecular mass of the compound having one phosphorus atom per molecule is :–

(A) 31 (B) 3.1×10^3 (C) 3.1×10^5 (D) 3.1×10^4 **MC0112**

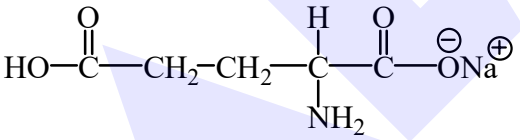
33. 13.4 gm of a sample of unstable hydrated salt : $Na_2SO_4.xH_2O$ was strongly heated. Weight loss on heating is found to be equal to 6.3 gm. Calculate the value of x.

(A) 6 (B) 5 (C) 7 (D) 8 **MC0113**

34. An organic compound contains 4% sulphur by mass. Its minimum molecular weight is :

(A) 200 (B) 400 (C) 800 (D) 1600 **MC0114**

35. Monosodium glutamate (MSG) is salt of one of the most abundant naturally occuring non-essential amino acid which is commonly used in food products like in **"MAGGI"** having structural formula as



Mass % of Na in MSG is-

(A) 14.8 (B) 15.1 (C) 13.6 (D) 16.5 **MC0115**

36. Which of the following series of compounds have same mass percentage of carbon ?

(A) CO_2, CO (B) CH_4, C_2H_6,C_2H_2

(C) C_2H_2, C_6H_6,$C_{10}H_8$ (D)HCHO, CH_3COOH,$C_6H_{12}O_6$ **MC0116**

37. A compound contains 69.5% oxygen and 30.5% nitrogen and its molecular weight is 92. The formula of that compound is :-

(A) N_2O (B) NO_2 (C) N_2O_4 (D) N_2O_5 **MC0117**

38. 1 lt. of a hydrocarbon weighs as much as one litre of CO_2 under identical conditions. The molecular formula of the hydrocarbon is -

(A) C_3H_8 (B) C_2H_6 (C) C_2H_4 (D) C_3H_6 **MC0118**

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39. Which of the following compounds has same empirical formula as that of glucose:-

(A) CH_3CHO (B) CH_3COOH (C) CH_3OH (D) C_2H_6 **MC0119**

40. Two oxides of a metal contains 50% and 40% of a metal respectively. The formula of the first oxide is MO.Then the formula of the second oxide is -

(A) MO_2 (B) M_2O_3 (C) M_2O (D) M_2O_5 **MC0120**

41. A compound of X and Y has equal mass of them. If their atomic weights are 30 and 20 respectively. The molecular formula of compound is -

(A) X_2Y_2 (B) X_3Y_3 (C) X_2Y_3 (D) X_3Y_2 **MC0121**

EUDIOMETRY

42. 10 ml CH_4 gas is burnt completely in air (O_2 = 20%, by volume). The minimum volume of air needed is -

(A) 20 ml (B) 50 ml (C) 80 ml (D) 100 ml **MC0122**

43. $C_6H_5OH\ (g) + O_2\ (g) \longrightarrow CO_2\ (g) + H_2O\ (l)$

Magnitude of volume change if 30 ml of C_6H_5OH (g) is burnt with excess amount of oxygen, is

(A) 30 ml (B) 60 ml (C) 20 ml (D) 10 ml **MC0123**

44. A mixture of C_2H_2 and C_3H_8 occupied a certain volume at 80 mm Hg. The mixture was completely burnt to CO_2 and H_2O(l). The pressure of CO_2 was found to be 230 mm Hg at the same temperature and volume. The fraction of C_2H_2 in mixture is

(A) 0.125 (B) 0.5 (C) 0.85 (D) 0.25 **MC0124**

45. 20 mL of a mixture of CO and H_2 were mixed with excess of O_2 and exploded & cooled. There was a volume contraction of 23 mL. All volume measurements corresponds to room temperature (27°C) and one atmospheric pressure. Determine the volume ratio $V_1 : V_2$ of CO and H_2 in the original mixture

(A) 6.5 : 13.5 (B) 5 : 15 (C) 9 : 11 (D) 7 : 13 **MC0125**

46. Each volume of a gaseous organic compound containing C, H and S only produce 1 volume CO_2, 2 volume H_2O vapours and 1 volume SO_2 gases on complete combustion. The molecular formula of compound is -

(A) CH_2S (B) CH_4S (C) C_2H_4S (D) C_2H_6S **MC0126**

47. For a chemical reaction occuring at constant pressure and temperature.

$$2A(g) + 5B(g) \longrightarrow C(g) + 2D(g)$$

(A) contraction in volume is double the volume of A taken if B is taken in excess.

(B) contraction in volume is more than the volume of B taken if A is in excess.

(C) volume contracts by 20 mL if 10 mL A is reacted with 20 mL B.

(D) no change in volume due to reaction **MC0127**

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48. When one litre of CO_2 is passed over hot coke, the volume becomes 1.4 litres. The composition of final products will not be.

(A) $V_{CO_2} : V_{CO} = 3:4$ (B) $V_{CO} = 1.6\,ltr.$ (C) $n_{CO_2} : n_{CO} = 3:4$ (D) % V of CO $= \frac{400}{7}$

MC0128

49. 10 ml of a compound containing 'N' and 'O' is mixed with 30 ml of H_2 to produce H_2O (l) and 10 ml of N_2 (g). Molecular formula of compound if both reactants reacts completely, is

(A) N_2O (B) NO_2 (C) N_2O_3 (D) N_2O_5 **MC0129**

50. When a definite volume of a gaseous alkyne (C_nH_{2n-2}) is burnt completely in excess of air, a contraction in volume equal to twice the volume of alkyne burnt occured. The value of 'n' is -

(A) 4 (B) 6 (C) 3 (D) 20 **MC0130**

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EXERCISE # O-II

SINGLE CORRECT :

1. A sample of iron ore, weighing 0.700g, is dissolved in nitric acid. The solution is then diluted with water, following with sufficient concentrated aqueous ammonia, to quantitative precipitation the iron as $Fe(OH)_3$. The precipitate is filtered, ignited and weighed as Fe_2O_3. If the mass of the ignited and dried precipitate is 0.541g, what is the mass percent of iron in the original iron ore sample (Fe = 56)

(A) 27.0 % (B) 48.1 % (C) 54.1 % (D) 81.1 % **MC0131**

2. A sample of pure Cu (4.00g) heated in a stream of oxygen for some time, gains in weight with the formation of black oxide of copper (CuO). The final mass is 4.90 g. What percent of copper remains unoxidized (Cu = 64)

(A) 90 % (B) 10 % (C) 20 % (D) 80 % **MC0132**

3. 1120 ml of ozonised oxygen ($O_2 + O_3$) at 1 atm & 273K weighs 1.76 gm. The reduction in volume on passing this through alkaline pyrogallol solution is -

(A) 896 ml (B) 224 ml (C) 448 ml (D) 672 ml **MC0133**

ASSERTION REASON:

4. **Statement-1 :** When a gaseous hydrocarbon is burnt in excess of oxygen and the products of combustion are cooled to the orignal temperatrue and pressure, a contraction in volume occurs.

Statement-2 : The contraction in volume is solely due to the liquifaction of water vapour.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true. **MC0134**

5. **Statement -1** : $2A + 3B \longrightarrow C$

4/3 moles of 'C' are always produced when 3 moles of 'A' & 4 moles of 'B' are added.

Statement -2 : 'B' is the limiting reactant for the given data.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is false, statement-2 is true.

(D) Statement-1 is true, statement-2 is false. **MC0135**

6. **Assertion** : During a chemical reaction, total moles remains constant.

Reason : During a chemical reaction, total mass remains constant.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true. **MC0136**

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MULTIPLE CORRECT :

7. 40 gm of a carbonate of an **alkali metal** or **alkaline earth metal** containing some inert impurities was made to react with excess HCl solution. The liberated CO_2 occupied 12.315 lit. at 1 atm & 300 K. The correct option is

(A) Mass of impurity is 1 gm and metal is Be (B) Mass of impurity is 3 gm and metal is Li

(C) Mass of impurity is 5 gm and metal is Be (D) Mass of impurity is 2 gm and metal is Mg

MC0137

8. 1 mole of H_2SO_4 will exactly neutralise :

(A) 2 mole of ammonia (B) 1 mole of $Ba(OH)_2$

(C) 0.5 mole of $Ca(OH)_2$ (D) 2 mole of KOH **MC0138**

9. 12 g of Mg was burnt in a closed vessel containing 32 g oxygen. Which of the following is /are correct.

(A) 2 gm of Mg will be left unburnt.

(B) 0.75 gm-molecule of O_2 will be left unreacted.

(C) 20 gm of MgO will be formed.

(D) The mixture at the end will weight 44 g. **MC0139**

10. 50 gm of $CaCO_3$ is allowed to react with 68.6 gm of H_3PO_4 then select the correct option(s)-

$$3CaCO_3 + 2H_3PO_4 \rightarrow Ca_3(PO_4)_2 + 3H_2O + 3CO_2$$

(A) 51.67 gm salt is formed (B) Amount of unreacted reagent = 35.93 gm

(C) n_{CO_2} formed = 0.5 moles (D) n_{H_2O} formed = 0.7 mole **MC0140**

11. Select the correct statement(s) for $(NH_4)_3PO_4$.

(A) Ratio of number of oxygen atoms to number of hydrogen atoms is 1 : 3

(B) Ratio of number of cations to number of anions is 3 : 1

(C) Ratio of number of gm-atoms of nitrogen to gm-atoms of oxygen is 3 : 2

(D) Total number of atoms in one mole of $(NH_4)_3PO_4$ is 20. **MC0141**

12. Two gases A and B which react according to the equation

$$aA_{(g)} + bB_{(g)} \longrightarrow cC_{(g)} + dD_{(g)}$$

to give two gases C and D are taken (amount not known) in an Eudiometer tube (operating at a constant Pressure and temperature) to cause the above.

If on causing the reaction there is no volume change observed then which of the following statement is/are correct.

(A) (a + b) = (c + d)

(B) average molecular mass may increase or decrease if either of A or B is present in limited amount.

(C) Vapour Density of the mixture will remain same throughout the course of reaction.

(D) Total moles of all the component of mixture will change. **MC0142**

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13. 100 ml mixture of CO and CO_2 mixed with 30 mL of O_2 and sparked in eudiometer tube. The residual gas after treatment with aq. KOH has a volume of 10 mL which remains unchanged when treated with alkaline pyrogallol. If all the volumes are under the same conditions, point out **correct** options(s):

(A) The volume of CO that reacts, is 60 mL

(B) The volume of CO that remains unreacted, is 10 mL

(C) The volume of O_2 that remains unreacted, is 10 mL

(D) The volume of CO_2 that gets absorbed by aq.KOH, is 90 mL. **MC0143**

Paragraph for Q.14 to Q.15

For the given series of reaction

$$4NH_3\ (g) + 5O_2\ (g) \longrightarrow 4NO\ (g) + 6H_2O\ (l)$$

$$2NO\ (g) + O_2\ (g) \longrightarrow 2NO_2\ (g)$$

14. If 20 ml of NH_3 is mixed with 100 ml of O_2. Volume contraction at the completion of above reactions is

(A) 20 ml (B) 85 ml (C) 35 ml (D) 100 ml **MC0144**

15. Total volume of O_2 used if 20 ml NH_3 is mixed with 100 ml O_2

(A) 40 (B) 60 (C) 35 (D) None of these

MC0144

Paragraph for Q.16 to Q.18

NaBr, used to produce AgBr for use in photography can be self prepared as follows :

$Fe + Br_2 \longrightarrow FeBr_2$ ....(i)

$FeBr_2 + Br_2 \longrightarrow Fe_3Br_8$ ....(ii) (not balanced)

$Fe_3Br_8 + Na_2CO_3 \longrightarrow NaBr + CO_2 + Fe_3O_4$ ....(iii) (not balanced)

(At. mass : Fe = 56, Br = 80)

16. Mass of iron required to produce 2.06×10^3 kg NaBr

(A) 420 gm (B) 420 kg (C) 4.2×10^5 kg (D) 4.2×10^8 gm

MC0145

17 If the yield of (ii) is 60% & (iii) reaction is 70% then mass of iron required to produce 2.06×10^3 kg NaBr

(A) 10^5 kg (B) 10^5 gm (C) 10^3 kg (D) None **MC0146**

18 If yield of (iii) reaction is 90% then mole of CO_2 formed when 2.06×10^3 gm NaBr is formed

(A) 20 (B) 10 (C) 9 (D) 440 **MC0147**

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Comprehension Q.19 and Q.20 (2 questions)

Estimation of halogens (Carius method) : A known mass of compound is heated with conc. HNO_3 in the presence of $AgNO_3$ contained in a hard glass tube known as carius tube in a furnce. C and H are oxidised to CO_2 and H_2O. The halogen forms the corresponding AgX. It is filtered, dried, and weighed.

Estimation of sulphur : A known mass of compound is heated with fuming HNO_3 or sodium peroxide (Na_2O_2) in the presence of $BaCl_2$ solution in Carius tube. Sulphur is oxidised to H_2SO_4 and precipitated as $BaSO_4$. It is filerted, dried and weighed.

19 0.15gm of an organic compound gave 0.12 gm of silver bromide by the Carius method. Find the percentage of bromine in the compound. (Ag = 108, Br = 80)

(A) 34.0 (B) 46.0 (C) 80.0 (D) 50.0 **MC0148**

20 0.32 gm of an organic substance when treated by Carius method gave 0.466 gm of $BaSO_4$. Calculate the percentage of sulphur in the compound. (Ba = 137)

(A) 10.0 (B) 34.0 (C) 20.0 (D) 30.0 **MC0148**

Comprehension Q.21 and Q.22 (2 questions)

Estimation of phosphorous : A known mass of compound is heated with fuming HNO_3 or sodium peroxide (Na_2O_2) in Carius tube which converts phosphorous to H_3PO_4. Magnesia mixture ($MgCl_2$ + NH_4Cl) is then added, which gives the precipitate of magnesium ammonium phosphate ($MgNH_4.PO_4$) which on heating gives magnesium pyrophosphate ($Mg_2P_2O_7$), which is weighed.

21 0.124 gm of an organic compound containing phosphorus gave 0.222 gm of $Mg_2P_2O_7$ by the usual analysis. Calculate the percentage of phosphorous in the compound.(Mg = 24, P = 31)

(A) 25 (B) 75 (C) 62 (D) 50 **MC0149**

22 An organic compound has 6.2 % of phosphorus. On sequence of reaction, the phosphorous present in the 10 gm of organic compound is converted to $Mg_2P_2O_7$. Find the weight of $Mg_2P_2O_7$ formed.

(A) 2.22 gm (B) 10.0 gm (C) 4.44 gm (D) 1.11 gm **MC0149**

TABLE TYPE :

	Column-I		**Column-II**		**Column-III**
(P)	60 gram sample of hydro carbon that contain 20% H and rest C	(1)	%C = 40	(i)	No. of atoms of C as well as O = $8N_A$
(Q)	240 gram urea	(2)	%H = $\frac{20}{3}$	(ii)	No. of C atoms = $4N_A$
(R)	120 gram acetic acid	(3)	% O = $\frac{160}{3}$	(iii)	No. of total atoms = $16N_A$
(S)	120 gram glucose	(4)	% N =46.7	(iv)	No. of total atoms is 2 times of no. of H atom

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23. Out of below correct matching is -

(A) P – 1 – i (B) P – 1 – ii (C) Q – 2 – iii (D) S – 2 – iv **MC0150**

24. In which of following is incorrect -

(A) Q – 2 – iv (B) R – 3 – iv (C) P – 4 – iii (D) R – 1 – ii **MC0150**

25. Out of below correct matching is -

(A) S – 4 – iv (B) R – 1 – ii (C) P – 4 – iii (D) P – 2 – ii **MC0150**

	Column-I (Gas taken)		**Column-II (O_2 needed for complete combustion)**		**Column-III (Contraction in volume)**
(1)	20 ml C_2H_4	(i)	60 ml	(I)	50 ml
(2)	25 ml C_3H_4	(ii)	100 ml	(II)	40 ml
(3)	30 ml C_2H_6	(iii)	70 ml	(III)	75 ml
(4)	35 ml CH_4	(iv)	105 ml	(IV)	70 ml

All volumes are measured at 25ºC and 1 atm.

26. Which of the following is correct match -

(A) 1 – i – II (B) 1 – iii – IV (C) 2 – iv – II (D) 2 – ii – III **MC0151**

27. Which of the following is correct match -

(A) 3 – iii – III (B) 3 – iv – III (C) 4 – iii – III (D) 4 – iv – IV **MC0151**

28. Which of the following is incorrect **(One or more than one correct)**

(A) 2 – ii – I (B) 4 – iii – IV (C) 3 – iv – IV (D) 1 – iii – II **MC0151**

MATCH THE COLUMN :

29. One type of artifical diamond (commonly called YAG for yttrium aluminium garnet) can be represented by the formula $Y_3Al_5O_{12}$. **[Y = 89, Al =27]**

Column I Element	**Column II Weight percentage**
(A) Y	(P) 22.73%
(B) Al	(Q) 32.32%
(C) O	(R) 44.95%

MC0152

30. The recommended daily dose is 17.6 milligrams of vitamin C (ascorbic acid) having formula $C_6H_8O_6$. Match the following. Given : $N_A = 6 \times 10^{23}$

Column I	**Column II**
(A) O-atoms present in daily dose	(P) 10^{-4} mole
(B) Moles of vitamin C in 1 gm of vitamin C	(Q) 5.68×10^{-3}
(C) Moles of vitamin C that should be consumed daily	(R) 3.6×10^{20}

MC0153

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31. Gaseous alkane (C_nH_{2n+2}) exploded with oxygen. Ratio of the mol of O_2 for complete combustion to the mole of CO_2 formed is given in column-I & in column II formula is given.

Column-I	**Column-II**
(A) 7 : 4	(P) C_3H_8
(B) 2 : 1	(Q) C_4H_{10}
(C) 5 : 3	(R) C_2H_6
(D) 13 : 8	(S) CH_4

MC0154

MATCHING LIST TYPE :

32.

Column-I	**Column-II (Mass of product)**
(P) $2H_2 + O_2 \rightarrow 2H_2O$ 1g 1g	(1) 1.028 g
(Q) $3H_2 + N_2 \rightarrow 2NH_3$ 1g 1g	(2) 1.333 g
(R) $H_2 + Cl_2 \rightarrow 2HCl$ 1g 1g	(3) 1.125 g
(S) $2H_2 + C \rightarrow CH_4$ 1g 1g	(4) 1.214 g

Code:

	P	**Q**	**R**	**S**
(A)	3	4	1	2
(B)	2	4	1	3
(C)	4	3	1	2
(D)	2	3	1	4

MC0155

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EXERCISE # JEE-MAINS

1. A gaseous hydrocarbon gives upon combustion 0.72 g of water and 3.08 g of CO_2. The empirical formula of the hydrocarbon is **[JEE(Main)-2013]**

(1) C_2H_4 (2) C_3H_4 (3) C_6H_5 (4) C_7H_8 **MC0175**

2. The ratio of masses of oxygen and nitrogen in a particular gaseous mixture is 1 : 4. The ratio of number of their molecule is : **[JEE(Main)-2014]**

(1) 1 : 8 (2) 3 : 16 (3) 1 : 4 (4) 7 : 32 **MC0174**

3. In Carius method of estimation of halogens, 250 mg of an organic compound gave 141 mg of AgBr. The percentage of bromine in the compound is :

(Atomic mass Ag = 108; Br = 80) **[JEE(Main)-2015]**

(1) 48 (2) 60 (3) 24 (4) 36 **MC0173**

4. The most abundant elements by mass in the body of a healthy human adult are :

Oxygen (61.4%) ; Carbon (22.9%), Hydrogen (10.0%) ; and Nitrogen (2.6%). The weight which a 75 kg person would gain if all ^1H atoms are replaced by ^2H atoms is **[JEE(Main)-2017]**

(1) 15 kg (2) 37.5 kg (3) 7.5 kg (4) 10 kg **MC0171**

5. 1 gram of a carbonate (M_2CO_3) on treatment with excess HCl produces 0.01186 mole of CO_2. the molar mass of M_2CO_3 in g mol^{-1} is :- **[JEE(Main)-2017]**

(1) 1186 (2) 84.3 (3)118.6 (4) 11.86 **MC0172**

6. The ratio of mass percent of C and H of an organic compound ($C_XH_YO_Z$) is 6 : 1. If one molecule of the above compound ($C_XH_YO_Z$) contains half as much oxygen as required to burn one molecule of compound C_XH_Y completely to CO_2 and H_2O. The empirical formula of compound $C_XH_YO_Z$ is **[JEE(Main)-2018 (offline)]**

(1) C_2H_4O (2) $C_3H_4O_2$ (3) $C_2H_4O_3$ (4) $C_3H_6O_3$ **MC0168**

7. For per gram of reactant, the maximum quantity of N_2 gas is produced in which of the following thermal decomposition reactions ? **[JEE(Main)-2018 (online)]**

(Given : Atomic wt. – Cr = 52u, Ba = 137u)

(1) $2NH_4NO_3(s) \rightarrow 2N_2(g) + 4H_2O(g) + O_2(g)$

(2) $Ba(N_3)_2(s) \rightarrow Ba(s) + 3N_2(g)$

(3) $(NH_4)_2Cr_2O_7(s) \rightarrow N_2(g) + 4H_2O(g)$

(4) $2NH_3(g) \rightarrow N_2(g) + 3H_2(g)$ **MC0169**

8. An unknown chlorohydrocarbon has 3.55% of chlorine. If each molecule of the hydrocarbon has one chlorine atom only; chlorine atoms present in 1 g of chlorohydrocarbon are :

(Atomic wt. of Cl = 35.5 u; Avogadro constant = 6.023×10^{23} mol^{-1}) **[JEE(Main)-2018 (online)]**

(1) 6.023×10^{21} (2) 6.023×10^{23} (3) 6.023×10^{20} (4) 6.023×10^{9} **MC0170**

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9. A 10 mg effervescent tablet containing sodium bicarbonate and oxalic acid releases 0.25 ml of CO_2 at T = 298.15 K and p = 1 bar. If molar volume of CO_2 is 25.0 L under such condition, what is the percentage of sodium bicarbonate in each tablet ? [Molar mass of $NaHCO_3$ = 84 g mol^{-1}]

[JEE(Main)-2019 (Jan.)]

(1) 16.8 (2) 8.4 (3) 0.84 (4) 33.6 **MC0160**

10. For the following reaction, the mass of water produced from 445 g of $C_{57}H_{110}O_6$ is :

$2C_{57}H_{110}O_6(s) + 163O_2(g) \rightarrow 114CO_2(g) + 110\ H_2O(l)$ **[JEE(Main)-2019 (Jan.)]**

(1) 495 g (2) 490 g (3) 890 g (4) 445 g **MC0161**

11. The percentage composition of carbon by mole in methane is : **[JEE(Main)-2019 (-april)]**

(1) 80% (2) 25% (3) 75% (4) 20% **MC0162**

12. For a reaction, **[JEE(Main)-2019 (-april)]**

$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$;

identify dihydrogen (H_2) as a limiting reagent in the following reaction mixtures.

(1) 14g of N_2 + 4g of H_2 (2) 28g of N_2 + 6g of H_2

(3) 56g of N_2 + 10g of H_2 (4) 35g of N_2 + 8g of H_2 **MC0163**

13. 5 moles of AB_2 weigh 125×10^{-3} kg and 10 moles of A_2B_2 weigh 300×10^{-3} kg. The molar mass of A(M_A) and molar mass of B(M_B) in kg mol^{-1} are : **[JEE(Main)-2019 (April)]**

(1) $M_A = 50 \times 10^{-3}$ and $M_B = 25 \times 10^{-3}$ (2) $M_A = 25 \times 10^{-3}$ and $M_B = 50 \times 10^{-3}$

(3) $M_A = 5 \times 10^{-3}$ and $M_B = 10 \times 10^{-3}$ (4) $M_A = 10 \times 10^{-3}$ and $M_B = 5 \times 10^{-3}$

MC0164

14. The minimum amount of $O_2(g)$ consumed per gram of reactant is for the reaction :

(Given atomic mass : Fe = 56, O = 16, Mg = 24, P = 31, C = 12, H = 1)

(1) $C_3H_8(g) + 5\ O_2(g) \rightarrow 3\ CO_2(g) + 4\ H_2O(l)$ **[JEE(Main)-2019 (April)]**

(2) $P_4(s) + 5\ O_2(g) \rightarrow P_4O_{10}(s)$

(3) $4\ Fe(s) + 3\ O_2(g) \rightarrow 2\ Fe_2O_3(s)$

(4) $2\ Mg(s) + O_2(g) \rightarrow 2\ MgO(s)$ **MC0165**

15. At 300 K and 1 atmospheric pressure, 10 mL of a hydrocarbon required 55 mL of O_2 for complete combustion and 40 mL of CO_2 is formed. The formula of the hydrocarbon is :

[JEE(Main)-2019 (Jan)]

(1) C_4H_8 (2) C_4H_7Cl (3) C_4H_{10} (4) C_4H_6 **MC0166**

16. 25 g of an unknown hydrocarbon upon burning produces 88 g of CO_2 and 9 g of H_2O. This unknown hydrocarbon contains. **[JEE(Main)-2019 (Jan)]**

(1) 20g of carbon and 5 g of hydrogen (2) 24g of carbon and 1 g of hydrogen

(3) 18g of carbon and 7 g of hydrogen (4) 22g of carbon and 3 g of hydrogen

MC0167

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17. $NaClO_3$ is used, even in spacecrafts, to produce O_2. The daily consumption of pure O_2 by a person is 492L at 1 atm, 300K. How much amount of $NaClO_3$, in grams, is required to produce O_2 for the daily consumption of a person at 1 atm, 300 K ? **[JEE(Main)-2020 (Jan)]**

$NaClO_3(s) + Fe(s) \rightarrow O_2(g) + NaCl(s) + FeO(s)$ R = 0.082 L atm mol^{-1} K^{-1} **MC0156**

18. 5 g of zinc is treated separately with an excess of **[JEE(Main)-2020 (Jan)]**

(a) dilute hydrochloric acid and

(b) aqueous sodium hydroxide.

The ratio of the volumes of H_2 evolved in these two reactions is :

(1) 1 : 4 (2) 1 : 2 (3) 2 : 1 (4) 1 : 1 **MC0157**

19. Amongst the following statements, that which was not proposed by Dalton was :

(1) all the atoms of a given element have identical properties including identical mass. Atoms of different elements differ in mass.

(2) chemical reactions involve reorganisation of atoms. These are neither created nor destroyed in a chemical reaction.

(3) when gases combine or reproduced in a chemical reaction they do so in a simple ratio by volume provided all gases are at the same T & P. **[JEE(Main)-2020 (Jan)]**

(4) matter consists of indivisible atoms. **MC0159**

20. A 20.0 mL solution containing 0.2 g impure H_2O_2 reacts completely with 0.316 g of $KMnO_4$ in acid solution. The purity of H_2O_2 (in %) is _____ (mol. wt. of H_2O_2 = 34; mol. wt. of $KMnO_4$ = 158)

[JEE(Main)-2020]

MC0212

21. The formula of a gaseous hydrocarbon which requires 6 times of its own volume of O_2 for complete oxidation and produces 4 times its own volume of CO_2 is C_xH_y. The value of y is _____.

[JEE(Main)-2021]

MC0213

22. Complete combustion of 1.80 g of an oxygen containing compound ($C_xH_yO_z$) gave 2.64 g of CO_2 and 1.08 g of H_2O. The percentage of oxygen in the organic compound is: **[JEE(Main)-2021]**

(1) 51.63 (2) 63.53 (3) 53.33 (4) 50.33 **MC0214**

23. The number of significant figures in 50000.020×10^{-3} is _____. **[JEE(Main)-2021]**

MC0215

24. Complete combustion of 750 g of an organic compound provides 420 g of CO_2 and 210 g of H_2O. The percentage composition of carbon and hydrogen in organic compound is 15.3 and _____ respectively. (Round off to the Nearest Integer) **[JEE(Main)-2021]**

MC0216

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25. The number of chlorine atoms in 20 mL of chlorine gas at STP is_____10^{21}. (Round off to the Nearest Integer). **[JEE(Main)-2021]**

[Assume chlorine is an ideal gas at STP

R = 0.083 L bar mol^{-1} K^{-1}, N_A = 6.023 $\times 10^{23}$] **MC0217**

26. Complete combustion of 3 g of ethane gives x × 10^{22} molecules of water. The value of x is _____. (Round off to the Nearest Integer). **[JEE(Main)-2021]**

[Use : N_A = 6.023 × 10^{23}; Atomic masses in u : C : 12.0 ; O : 16.0 ; H : 1.0] **MC0218**

27. 4g equimolar mixture of NaOH and Na_2CO_3 contains x g of NaOH and y g of Na_2CO_3. The value of x is _____ g. (Nearest integer) **[JEE(Main)-2021]**

MC0219

28. When 0.15 g of an organic compound was analyzed using Carius method for estimation of bromine, 0.2397 g of AgBr was obtained. The percentage of bromine in the organic compound is _____. (Nearest integer)

[Atomic mass : Silver = 108, Bromine = 80] **[JEE(Main)-2021]**

MC0220

29. Consider the complete combustion of butane, the amount of butane utilized to produce 72.0 g of water is _____ × 10-1 g. (in nearest integer) **[JEE(Main)-2021]**

MC0221

30. The number of significant figures in 0.00340 is _____ . **[JEE(Main)-2021]**

MC0222

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EXERCISE # JEE-ADVANCED

1. A student performs a titration with different burettes and finds titre values of 25.2 mL, 25.25 mL, and 25.0 mL. The number of significant figures in the average titre value is :- **[JEE 2010]**

MC0223

2. Silver (atomic weight = 108 g mol^{-1}) has a density of 10.5 g cm^{-3}. The number of silver atoms on a surface of area 10^{-12} m^2 can be expressed in scientific notation as y ´ 10^x. The value of x is

[JEE 2010]

MC0224

3. If the value of Avogadro number is 6.023×10^{23} mol^{-1} and the value of Boltzmann constant is 1.380×10^{-23} JK^{-1}, then the number of significant digits in the calculated value of the universal gas constant is **[JEE 2014]**

MC0180

4. The ammonia prepared by treating ammonium sulphate with calcium hydroxide is completely used by $NiCl_2.6H_2O$ to form a stable coordination compound. Assume that both the reactions are 100% complete. If 1584 g of ammonium sulphate and 952g of $NiCl_2.6H_2O$ are used in the preparation, the combined weight (in kg) of gypsum and the nickel-ammonia coordination compound thus produced is___. **[JEE 2018]**

$$(NH_4)_2SO_4 + Ca(OH)_2 \rightarrow CaSO_4.2H_2O + 2NH_3$$

$$NiCl_2 \cdot 6H_2O + 6NH_3 \rightarrow \left[Ni(NH_3)_6\right]Cl_2 + 6H_2O$$

(Atomic weights in g mol^{-1}: H = 1, N = 14, O = 16, S = 32, Cl = 35.5, Ca = 40, Ni = 59)

MC0178

5. Galena (an ore) is partially oxidized by passing air through it at high temperature. After some time, the passage of air is stopped, but the heating is continued in a closed furnance such that the contents undergo self-reduction. The weight (in kg) of Pb produced per kg of O_2 consumed is _____ .

$PbS + O_2 \longrightarrow Pb + SO_2$ **[JEE 2018]**

(Atomic weights in g mol^{-1} : O = 16, S = 32, Pb = 207) **MC0179**

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ANSWER KEY

EXERCISE # S- I

1. (i) 0.16 or 0.17 (ii) 3200.00
(iii) 23.38 (iv) (18.06 or 18.07) , (54.19 or 54.20)
(v) 12.04

2.	1.99	3. 3.20	4. 180.00	5. 2.43 or 2.44
6.	(9.48)	7. (55.00)	8. (34.05)	9. (2.00)
10.	(102.50)	11. 36.50	12. 27.60	13. 3.61
14.	0.63	15. 58.80	16. 6.41 or 6.42	17. 0.11 or 0.12
18.	0.25	19. 5.60	20. %Al = 60; %Mg = 40	

21. $\%CaCO_3 = 28.40$ or 28.41, $\%MgCO_3 = 71.59$

22. $\%NaHCO_3 = 16.80$; $\%Na_2CO_3 = 83.20$

23.	0.75	24. 1.70	25. 0.50	26. 39.18
27	4.48	28. 61.50	29. 0.20	30. (50.00)
31.	(31.42 or 31.43)	32. (0.64)	33. (40.00)	34. (0.72)
35.	(0.04)	36. (3.04)	37. 6.40	38. (a) 0.71 (b) 61.97
39.	1.20	40. (4.00)	41. (33.33)	42. 5.28
43.	(4.00)	44. (2.00)	45. (0.07)	46. (19.00)
47.	$C_6H_4Cl_2$	48. (3.00)	49. ($C_2H_4O_2$)	50. (0.50)
51.	140.00	52. 30.00	53. 70.00	54. (10.00)
55.	73.33	56. 10.00	57. (N_2O_5)	58. (C_2H_6)
59.	(3.00)	60. (5.00)		

EXERCISE # S-II

1.	(i) 0.5 , 0.5 ; (ii) 2, 1 (iii) 1, 2		2. (65.62)	3. 0.66
4	%NaCl = 77.80	5. (a) $Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2Fe$; (b) 2.96 ; (c) 10,000 units		
6.	5.74	7. Ans. 6.02	8. (1.50)	9. (222.00)
10.	(11.00)	11. (5.00)	12. (1.70)	13. (3.30)

14. % Ag = $\frac{0.5}{0.66} \times 100 = 75.75$

% Ag = $\frac{1}{1.32} \times 100 = 75.75$

In both AgCl, % Ag is same.

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15. $\% \text{ Pb} = \frac{6.488}{(6.488+1.002)} \times 100 = 86.62\%$

16. **Oxide = 1 gm ⇒ oxygen = 0.5 gm + element = 0.8 gm**

Ratio = 1 : 1

Oxide = 4 gm ⇒ oxygen = 3.2 gm + element = 0.8 gm

Ratio = 4 : 1

17. **0.11 gm oxide gives ⇒ 0.07 gm N , Hence ; 0.07 gm N + 0.04 gm O**

0.15 gm oxide gives ⇒ 0.07 gm N , Hence ; 0.07 gm N + 0.08 gm O

18. **CH_4 = 4.50, CO_2 = 1.50, C_2H_4 = 4.00** **19. (0.66 or 0.67)** **20. 9.00**

EXERCISE # O-I

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	D	C	C	C	A	B	C	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	C	B	B	A	C	B	A	B	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	D	B	A	A	D	A	A	B	A	A
Que.	31	32	33	34	35	36	37	38	39	40
Ans.	A	C	C	C	C	D	C	A	B	B
Que.	41	42	43	44	45	46	47	48	49	50
Ans.	C	D	B	A	D	B	A	B	C	C

EXERCISE # O-II

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	C	B	A	C	C	D	B	A,B,D	B,C,D	A,B,C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,B	A,C	A,B,D	C	C	B	C	B	A	C
Que.	21	22	23	24	25	26	27	28		
Ans.	D	A	D	C	B	A	B	C,D		
Que.	29			30			31			
Ans.	A-R	B-P	C-Q	A-R	B-Q	C-P	A-R	B-S	C-P	D-Q
Que.	32									
Ans.	A									

EXERCISE # JEE-MAINS

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	4	4	3	3	2	3	4	3	2	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	3	3	3	4	2	2120 to 2140	4	3	85
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	8	3	8	3	1	18	1	68	464	3

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	
Ans.	3	7	4	2.99	6.46 or 6.47	

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